

Aditya

Generative AI Engineer | LLM Systems | Agentic AI | NLP

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SUMMARY

- Design controllable autonomous systems around LLMs using structured state-driven workflows and context management.
- Solid foundation in Machine Learning, NLP, and applied mathematics (linear algebra, probability, statistics), with hands-on experience fine-tuning and evaluating Generative AI models.
- Focus on reliability-first architectures that reduce drift and stabilize reasoning.
- Build modular, interpretable agentic systems that balance autonomy with structured control.

SKILLS

Programming: Python, C++, C, SQL, Java

Frameworks: PyTorch, Hugging Face, LangChain, LangGraph, Flask

Generative AI: LLMs, Transformer Architectures (GPT-style), Fine-tuning & Pre-training, Prompt Engineering, LoRA, QLoRA

NLP: Text Preprocessing, Tokenization, Embeddings, Grammar/Text Correction, Retrieval-Augmented Generation (RAG)

Concepts: Machine Learning (Supervised & Unsupervised Learning), Deep Learning, Computer Vision Fundamentals (CNN architectures, object detection concepts), Multi-Agent Systems & Agentic AI, Model Optimization & Evaluation, Data Preprocessing & Feature Engineering

Mathematical Foundations: Linear Algebra, Probability & Statistics (B.Tech CS coursework)

PROJECTS

Context Hygiene System | *LangGraph, Agentic AI, Context Control* Jan 2026

- Designed a context hygiene layer for agentic systems to control long-term conversation memory and token growth.
- Implemented adaptive context pruning, history compression, and token-threshold monitoring to prevent context window overflow.
- Added drift detection and state-transition mechanisms using LangGraph to maintain structured conversational flow.
- Improved reasoning stability in multi-step agent workflows by preserving task-critical context while filtering noise.

Advanced Adaptive RAG System | *LangChain, ChromaDB, Web Scraping, Multi-LLM Support* Oct - Nov 2025

- Designed a multi-source ingestion pipeline supporting PDF uploads and dynamic web scraping, storing embeddings in ChromaDB.
- Applied text preprocessing and feature engineering (chunking, embedding generation) to optimize retrieval quality.
- Implemented hybrid retrieval (semantic + keyword-based) with LLM-based document grading for relevance filtering.
- Built a self-evaluating generation loop that classifies outputs as useful, not useful, or unsupported.
- Integrated fallback web search and regeneration mechanisms to improve answer completeness.
- Architected a modular LLM abstraction layer enabling seamless model switching.
- Reduced unsupported responses by introducing conditional fallback retrieval based on LLM output grading.

Advanced RAG with LCEL & Local LLM | *LangChain LCEL, Local LLMs, FAISS* Sep - Oct 2026

- Built a fully local RAG pipeline using LangChain Expression Language.
- Enabled private document QA without cloud dependency.

MediAgent System | *Multi-Agent AI, LLMs* Dec 2025

- Developed a multi-agent medical assistant for structured symptom analysis.
- Coordinated planner and executor agents for controlled reasoning.

GrammarFix - AI Grammar Correction | *Flask, T5, Gemini API, NLP* June 2025

- Built a hybrid NLP web app for real-time grammar correction, covering text preprocessing and model-based error classification.
- Integrated local and cloud LLMs for low-latency inference and deployed the app for production-style usage via Flask.

LLaMA-2 7B Fine-Tuning | *PyTorch, QLoRA, LoRA, Hugging Face* June 2025

- Fine-tuned LLaMA-2 7B using QLoRA on Google Colab for grammar error reasoning tasks.
- Trained on datasets from Hugging Face and Kaggle using parameter-efficient fine-tuning.

EDUCATION

Vivekananda Global University - B.Tech in Computer Science & Engineering

Aug 2023 - May 2027 | Jaipur, India | 4th Year (Current) | CGPA(9.0)

JM Sarvodaya Vidyalaya, Sec-12 - Higher Secondary Education

2021 | New Delhi, India | Percentage: 85%

CERTIFICATIONS

- Generative AI Engineering - IBM
- Agentic AI - Coursera
- Machine Learning - Coursera
- Deep Learning Specialization - DeepLearning.AI

ADDITIONAL INFORMATION

- Languages: English (Fluent), Japanese (Basic)
- Cricket: District Level; DDCA U-16 State Camp